

ME 6030 - Assignment 2

Theory Question 1

Let $P = A(A^\top A)^{-1}A^\top$ be the projection matrix onto the column space of A , where $A \in \mathbb{R}^{m \times n}$ has linearly independent columns.

(a) P is symmetric: $P^\top = P$

Proof. Taking the transpose and using $(XY)^\top = Y^\top X^\top$:

$$P^\top = (A(A^\top A)^{-1}A^\top)^\top = (A^\top)^\top ((A^\top A)^{-1})^\top A^\top = A((A^\top A)^\top)^{-1}A^\top.$$

Since $(A^\top A)^\top = A^\top A$, we get $P^\top = A(A^\top A)^{-1}A^\top = P$. □

(b) P is idempotent: $P^2 = P$

Proof.

$$P^2 = A(A^\top A)^{-1}A^\top \cdot A(A^\top A)^{-1}A^\top = A \underbrace{(A^\top A)^{-1}(A^\top A)}_{=I} (A^\top A)^{-1}A^\top = A(A^\top A)^{-1}A^\top = P.$$

□

(c) $I - P$ is a projection matrix

Proof. We verify the two properties of a projection matrix.

Symmetry: $(I - P)^\top = I^\top - P^\top = I - P$, using part (a).

Idempotency:

$$(I - P)^2 = I - 2P + P^2 = I - 2P + P = I - P,$$

using part (b).

Hence $I - P$ is a projection matrix. It projects onto the *left null space* (orthogonal complement of the column space) of A , i.e., $\text{null}(A^\top)$. To see this: for any v , $(I - P)v = v - Pv$, and Pv is in $\text{col}(A)$, so $(I - P)v$ is orthogonal to $\text{col}(A)$, placing it in $\text{col}(A)^\perp = \text{null}(A^\top)$. □

(d) Eigenvalues of P are only 0 and 1

Proof. Let $Pv = \lambda v$ for some eigenvector $v \neq 0$. Apply P to both sides:

$$P^2v = P(\lambda v) = \lambda Pv = \lambda^2v.$$

But from part (b), $P^2v = Pv = \lambda v$. Therefore $\lambda^2v = \lambda v$, which gives

$$\lambda^2 = \lambda \implies \lambda(\lambda - 1) = 0 \implies \lambda \in \{0, 1\}.$$

□

Theory Question 2

Given an inconsistent system $A\mathbf{x} = \mathbf{b}$ where $A \in \mathbb{R}^{m \times n}$, $m > n$.

(a) The least squares solution satisfies the normal equations $A^\top A\hat{\mathbf{x}} = A^\top \mathbf{b}$

Proof. The least squares problem minimises $f(\mathbf{x}) = \|A\mathbf{x} - \mathbf{b}\|^2$. Expanding:

$$f(\mathbf{x}) = (A\mathbf{x} - \mathbf{b})^\top (A\mathbf{x} - \mathbf{b}) = \mathbf{x}^\top A^\top A\mathbf{x} - 2\mathbf{b}^\top A\mathbf{x} + \mathbf{b}^\top \mathbf{b}.$$

Setting the gradient to zero:

$$\nabla_{\mathbf{x}} f = 2A^\top A\mathbf{x} - 2A^\top \mathbf{b} = 0 \implies A^\top A\hat{\mathbf{x}} = A^\top \mathbf{b}.$$

□

(b) The residual $\mathbf{r} = \mathbf{b} - A\hat{\mathbf{x}}$ is orthogonal to $\text{col}(A)$

Proof. Let $\mathbf{r} = \mathbf{b} - A\hat{\mathbf{x}}$. From the normal equations:

$$A^\top (\mathbf{b} - A\hat{\mathbf{x}}) = A^\top \mathbf{b} - A^\top A\hat{\mathbf{x}} = 0.$$

So $A^\top \mathbf{r} = 0$, which means $\mathbf{r}$ is orthogonal to every column of A , i.e., $\mathbf{r} \perp \text{col}(A)$. □

(c) If A has full column rank, $A^\top A$ is symmetric positive definite and the solution is unique

Proof. Symmetry: $(A^\top A)^\top = A^\top (A^\top)^\top = A^\top A$.

Positive definiteness: For any $\mathbf{x} \neq \mathbf{0}$:

$$\mathbf{x}^\top (A^\top A)\mathbf{x} = (A\mathbf{x})^\top (A\mathbf{x}) = \|A\mathbf{x}\|^2.$$

Since A has full column rank, $A\mathbf{x} = \mathbf{0}$ only when $\mathbf{x} = \mathbf{0}$. Therefore $\|A\mathbf{x}\|^2 > 0$ for all $\mathbf{x} \neq \mathbf{0}$, so $A^\top A$ is positive definite.

Uniqueness: Because $A^\top A$ is positive definite it is invertible, so the normal equations $A^\top A\hat{\mathbf{x}} = A^\top \mathbf{b}$ have a unique solution $\hat{\mathbf{x}} = (A^\top A)^{-1} A^\top \mathbf{b}$. □